Instructions:

Create a team of up to 5 people for solving short assignments.

Task:

Open EViews and the workfile "PRICES". It contains monthly data from 2000M01 to 2026M06:

- Harmonised index of consumer prices for goods, Latvia, 2025=100, seasonally adjusted ('lv').
- Harmonised index of consumer prices for goods, Lithuania, 2025=100, seasonally adjusted ('lt').

Latvia and Lithuania are small open economies, euro-area neighbours with similar consumption baskets, so goods-market arbitrage and common shocks should tie their goods prices together. The economic hypothesis is the law of one price: in the long run $\log(lv) = \beta_0 + \beta_1 \cdot \log(lt)$ with $\beta_1 = 1$, so the relative price log(lv/lt) is stationary.

The seminar asks three business decision-relevant questions:

- Does the law of one price hold in the long run?
- How far can the two countries' goods prices drift apart?
- How long does a gap last before it closes?

We will use the sample from 2010m1 to 2026m6 for solving the tasks (adjust the sample).

1. The long-run relationship (Engle-Granger, step 1)

a. Plot log(lv) and log(lt) together, and plot the relative price log(lv) − log(lt). Based on the charts, do you expect the two series to be cointegrated. Explain.

Hint: Look at the relative price from part (a): if it already looks stationary, what does that tell you about cointegration and about the likely value of β_1?

b. Confirm the order of integration of log(lv) and log(lt). Run ADF (and KPSS as a confirmatory check). Make a conclusion.

c. Estimate the static long-run regression by OLS: log(lv) on log(lt) and an intercept. Record β_1 and compare it with the law-of-one-price value. Should we test $\beta_1 = 1$ from this output?

Hint: Note why the OLS point estimate is acceptable under cointegration (super-consistency) but its standard errors are not yet trustworthy.

d. Test for cointegration. Save the OLS residuals from part (c), run an ADF on them, and compare the τ-statistic to the Engle–Granger critical values (not the standard ADF values EViews prints).

Hint: compute the 5% critical value (CV) for your actual T (number of observations) from: $CV = -3.3377 - 5.967/T - 8.98/T^2$.

Confirm with the EViews built-in test. State the conclusion.

Hint: Select log(lv) and log(lt) and open them as group. View → Cointegration Test → Single-Equation Cointegration Test → Engle–Granger. Look at tau-statistic and the corresponding p-value.

e. Re-estimate the long-run relationship efficiently by FMOLS and by DOLS. Compare β_1 across FMOLS, DOLS and OLS. Test the law of one price formally and make a conclusion. If it is not rejected, look at the residuals and answer the question: how far do the two countries' goods prices typically deviate?

Hint: Compare the slopes across OLS/FMOLS/DOLS, not the intercepts: the constant shifts mechanically with β_1 and its t-statistic is not a reliable test in a cointegrating regression. FMOLS/DOLS fix the standard errors (endogeneity + serial correlation), which is what lets you legitimately test $\beta_1 = 1$.

Discussion questions – Part 1

1. A regression of two trending price indices can show $R^2 \approx 0.99$ and a hugely significant slope yet not be cointegrated. What is that trap called, and what does the residual test add that R^2 and the *t*-statistic cannot?

2. If the law of one price holds exactly ($\beta_1 = 1$), what is the economic meaning of the constant β_0, and why should you be cautious reading it off indices rebased to 2025 = 100?

2. The error-correction model (Engle-Granger, step 2)

Proceed with the FMOLS long-run relationship from part (1.e).

a. Estimate the VECM to answer the question: how long does a gap last before it closes?

Hint: Create a VAR for the first log-differences of Latvian and Lithuanian prices. Add the lagged residuals from the long-run equation as an exogenous variable together with an intercept. Use the information criteria to choose the appropriate number of lags. Test the VECM for stationarity and check the residuals autocorrelation.

b. Check the significance and signs of the error-correction coefficients. Answer the question – who does the adjustment? Make the conclusion.

Hint: If one country's α is significant and the other's is not, that country is the price follower and the other is the anchor (weakly exogenous).

c. Based on the results from parts (a) and (b), can we proceed with the single-equation ECM? If the answer is yes, estimate the single-equation ECM by OLS.

Hint: Check lags of left- and right-hand side variables. Use impulse dummies for the energy crises episodes (2022–2023). Perform the model's diagnostics. Adjust standard errors if necessary.

d. Interpret the error-correction coefficient from part (c). Answer the question: how long does a gap last before it narrows to 10%?

Hint: The equilibrium error follows approximately $EC_t = (1 + \alpha)EC_{t-1}$. That is a geometric decay with factor $(1 + \alpha)$.

e (optional). As a robustness check, flip the normalization: re-estimate the long-run relationship as log(lt) on log(lv), rebuild the VECM, and confirm the who-adjusts verdict does not depend on which variable you put on the left.

Discussion questions – Part 2

1. Suppose the half-life comes out around six months. Translate that into a sentence a retailer or distributor operating in both countries would act on.
2. You found that one country adjusts and the other is weakly exogenous. Does that prove the follower's prices are 'caused' by the leader's, or could a third factor be driving both? Name one variable that could act as the common driver of both countries' goods prices?
3. The single-equation Engle–Granger ECM assumes the other country is weakly exogenous. What did estimating both equations let you check that a single equation could not?